Time Series Forecasting

Germany Monthly Electricity Demand (2016 – 2025)

1. Problem Definition

Germany is Europe's largest electricity market. Managing electricity supply reliably and efficiently requires grid operators, energy traders, and policymakers to anticipate demand in advance. Electricity cannot be economically stored at scale, meaning that supply and demand must be balanced continuously. Overestimating demand leads to costly surplus generation or unnecessary imports; underestimating it risks supply shortfalls and grid instability.

This analysis focuses on forecasting Germany's monthly electricity available to the internal market, measured in gigawatt-hours (GWh), using data sourced from Eurostat (dataset: NRG_CB_EM). The series spans January 2016 to November 2025, covering 119 monthly observations capturing pre-pandemic demand patterns, the COVID-19 shock of 2020, the energy crisis of 2021-2022 triggered by rising gas prices, and the structural decline in demand that followed.

Decision-maker

The primary decision-maker is a national grid operator such as Bundesnetzagentur (Germany's Federal Network Agency), or an energy trading desk at a large utility. These actors must plan generation capacity commitments, reserve requirements, and cross-border import/export contracts on a monthly horizon.

What decision depends on the forecast Monthly demand forecasts directly inform:

1. Capacity scheduling: how much generation capacity (coal, gas, renewables) to contract for each month
2. Reserve margins: how much backup capacity must be kept available
3. Energy procurement: decisions on importing electricity from neighbouring countries

Why forecasting is needed

Electricity demand is driven by temperature, economic activity, and structural efficiency trends, all of which evolve over time. A simple assumption of constant demand would fail to capture the strong winter peaks and summer troughs visible in the data (demand ranging from approximately 34,500 GWh to 44,800 GWh across months), nor the downward structural trend observed since 2022 due to energy efficiency improvements and industrial contraction. Formal time series forecasting allows these patterns to be modelled explicitly and extrapolated into future planning horizons.

2. Data Description and Pre-processing

- **Source:** Eurostat, dataset NRG_CB_EM- Supply, transformation and consumption of electricity, monthly data.
- **Series:** Electricity available to the internal market from January 2016 to November 2025, giving 119 monthly observations at monthly frequency
- **Descriptive statistics:** mean = 38209 GWh, standard deviation = 2482 GWh, minimum = 33692 GWh (April 2025), maximum = 44805.4 GWh (January 2017)
- There are no missing values, no transformations (e.g. log) were applied because the seasonal amplitude appear roughly constant in absolute terms, indicating an additive rather than multiplicative structure (confirmed in Section 3.1).

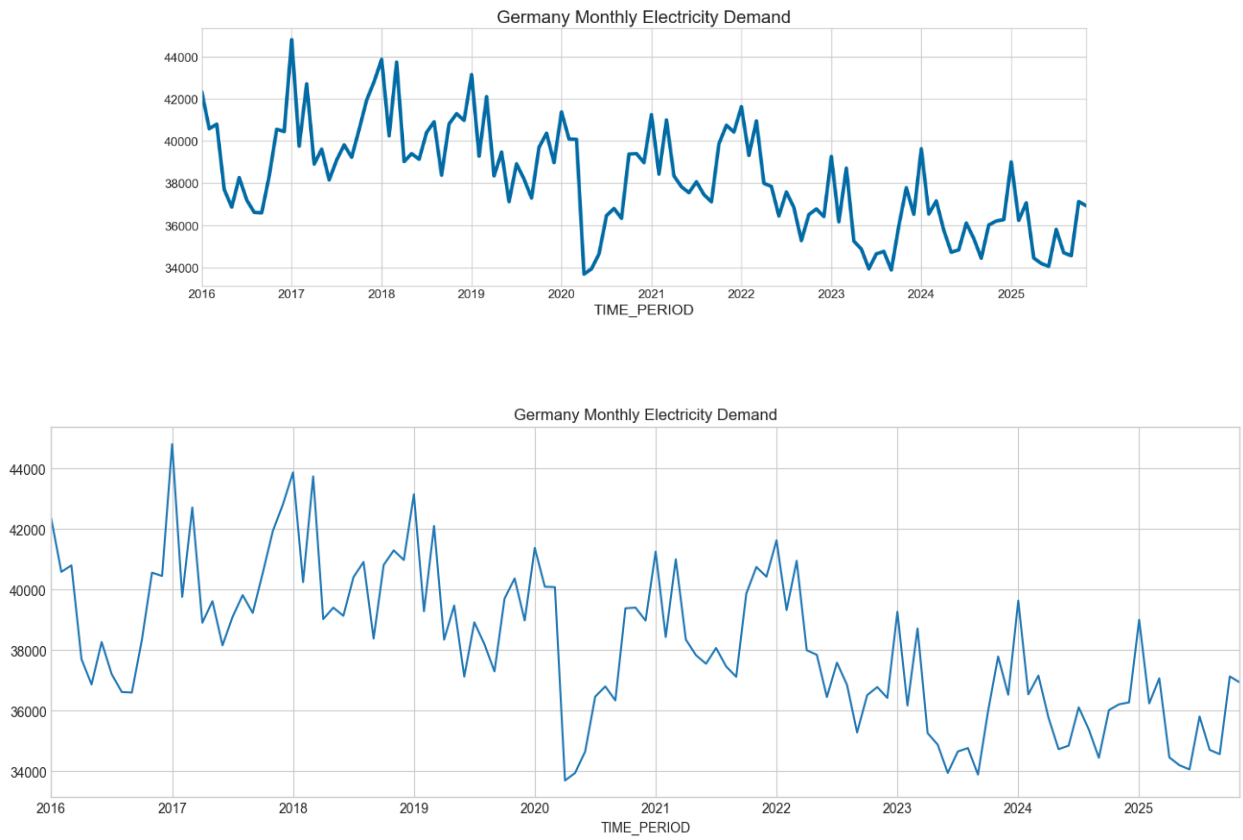


Figure 1. Germany monthly electricity available to the internal market (GWh), Jan 2016–Nov 2025.

3. Exploratory Analysis

3.1 Decomposition

To determine the appropriate structural for the electricity demand series, both additive and multiplicative decomposition method were tested. The additive model was selected because the seasonal fluctuations, the magnitude of the winter peak and summer dips remained relatively constant in absolute terms regardless of the trend. And we also test the statistical test of the residuals, where the additive model generated a more stable standard deviation and more symmetric distribution of errors.

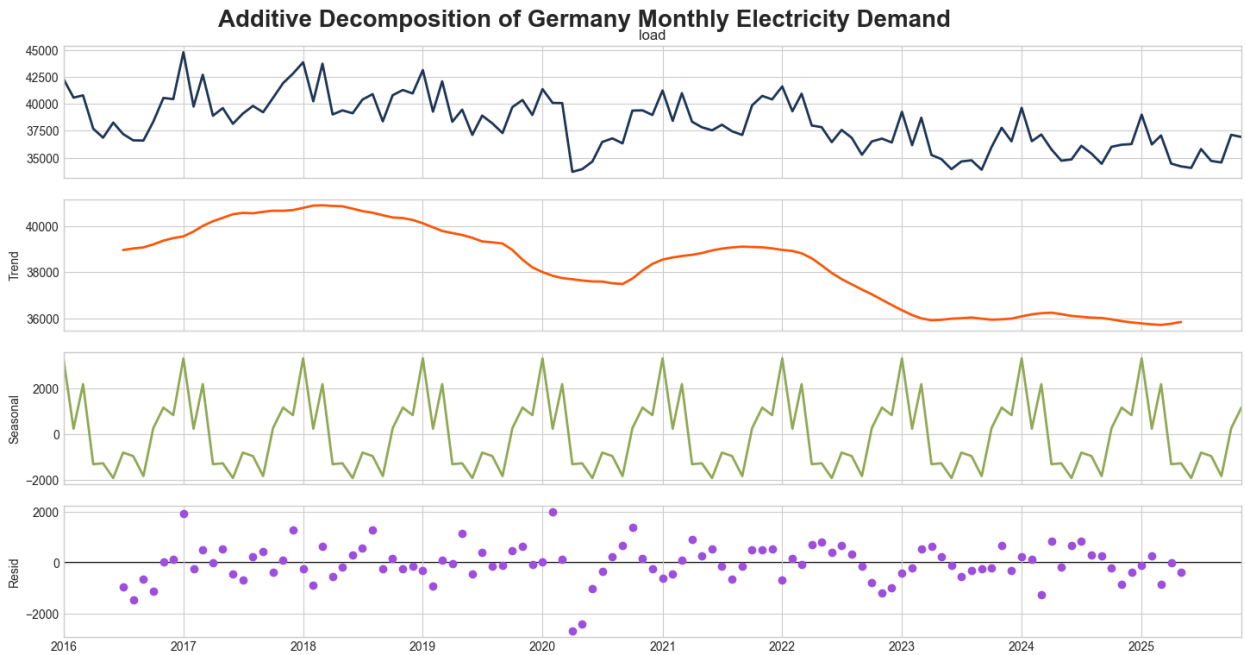
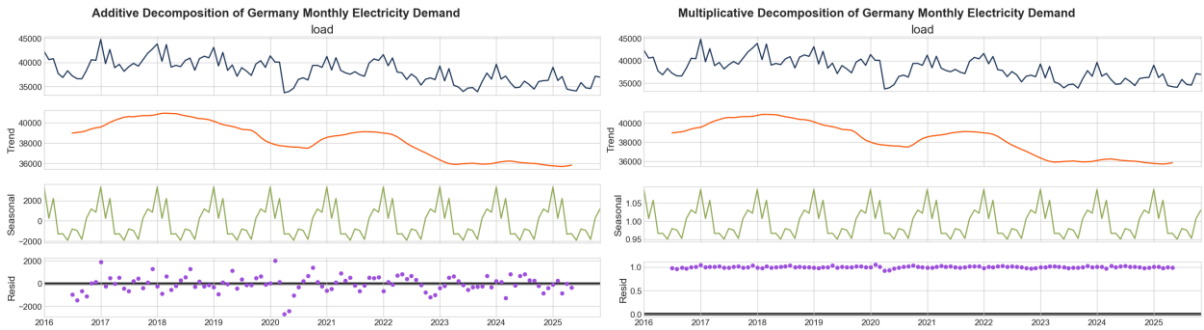


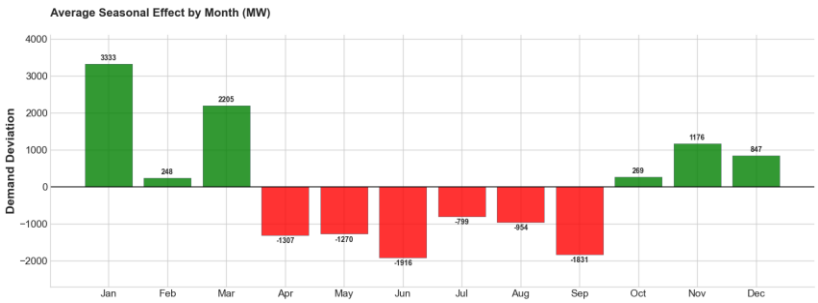
Figure 2. Additive decomposition (observed, trend, seasonal, residual) for Germany monthly electricity demand.



3.2 Trend, Seasonality, and Residuals

Trend: The trend component show a mild decline from around 42,000 GWh 2016-2018 to approximately 36,000 GWh by 2024-2025. No abrupt structural break is visible, though a light acceleration of the decline post-2020 may reflect COVID-19 demand suppression followed by structural change.

Seasonality: A stable annual cycle is present, with winter peaks (December–February) and summer troughs (June–August). The seasonal amplitude remains broadly constant over the full sample, supporting the additive specification.



Residuals: The residuals show no obvious pattern but have some spikes during 2020-2021, suggesting decomposition alone would be insufficient for forecasting. (Confirmed in Section 3.3)

3.3 ACF / PACF

To determine the model structure, the stationary test was applied. The original data was confirmed to be non-stationary (p=0.78), consistent with the visible trend in the decomposition. Applying first-order differencing effectively removed this trend, resulting in a stationary series (p =0.0027) While a further test on seasonal differencing yielded a borderline p-value of 0.054, this marginal result was identified as an effect of the 2020-2021 COVID-19 demand shock rather than a seasonal unit root. Visually, the series differenced at d=1 already fluctuates around a constant mean of zero with stable variance. Hence, a single order of regular differencing is sufficient to achieve stationarity, leading to a final model specification of d=1 and D=0.

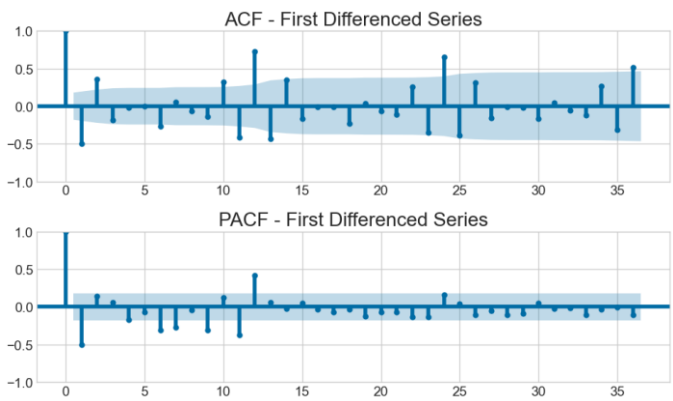
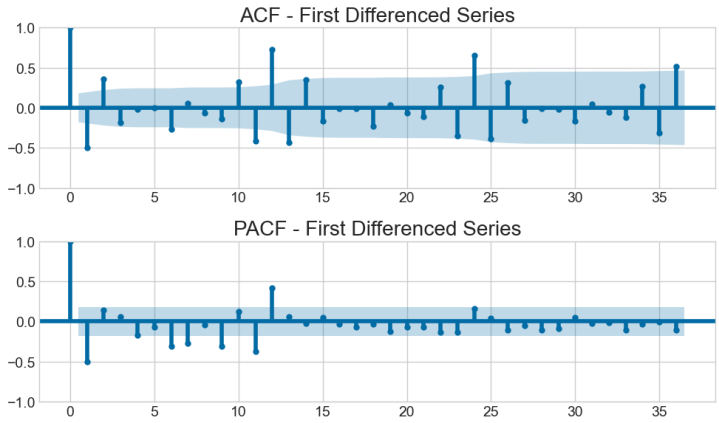


Figure 3. ACF and PACF of the first-differenced series.



4. Forecasting Models

4.1 Benchmark model: Seasonal naïve

The seasonal naïve method provides a strong benchmark for monthly data with annual seasonality. The forecast for month t is set equal to the observation from the same month in the previous year: $\hat{y}_t = y_{t-12}$. This method requires no parameter estimation and is appropriate when seasonality is stable and the level changes slowly. In this dataset, the annual winter–summer cycle is persistent, making the seasonal naïve forecast a meaningful baseline for comparison.

4.2 Exponential smoothing: Holt–Winters additive ETS

An additive Holt–Winters exponential smoothing model was fitted because (i) decomposition indicates additive seasonality (approximately constant seasonal amplitude in GWh), and (ii) the series exhibits a slowly evolving downward level/trend. The model updates three components each period: level, additive trend, and additive seasonal indices, using exponentially decaying weights controlled by smoothing parameters (α for level, β for trend, and γ for seasonality). This structure is well suited to demand series where the seasonal pattern persists but the underlying level drifts over time.

4.3 Non-seasonal ARIMA(1,1,0)

After first-order differencing ($d = 1$), the ACF/PACF of the differenced series suggests a short-memory autocorrelation structure at low lags. A parsimonious ARIMA(1,1,0) was therefore fitted as a non-seasonal baseline. While it can capture short-run persistence in the differenced series, it does not explicitly model seasonal dynamics and is expected to underperform when strong annual seasonality is present.

4.4 Seasonal ARIMA: SARIMA(1,1,0)(1,0,1)[12]

To explicitly represent annual seasonality ($s = 12$), a SARIMA(1,1,0)(1,0,1)[12] model was fitted. The $(p,d,q) = (1,1,0)$ component captures short-run dependence after removing the trend via differencing, while the seasonal AR and MA terms ($P = 1$, $Q = 1$) at lag 12 capture dependence between the same months in consecutive years. This model family is appropriate when both trend and seasonality are present and when residual diagnostics indicate remaining seasonal autocorrelation under a non-seasonal ARIMA.

5. Forecast Evaluation and Comparison

5.1 Evaluation design

Two complementary evaluation strategies were used. First, a single hold-out split was defined with training data from Jan 2016–Dec 2023 and a test period from Jan 2024–Nov 2025 (23 months). Second, to reduce dependence on any single split point, an expanding-window time series cross-validation was implemented with five folds, each using a 12-month forecast horizon. Models were compared using

two scale-dependent accuracy measures: mean absolute error (MAE) and root mean squared error (RMSE), both expressed in GWh.

5.2 Single train/test split results

Model	MAE (GWh)	RMSE (GWh)
Seasonal naïve	688.43	858.13
ETS (Holt–Winters additive)	765.72	892.02
ARIMA(1,1,0)	1,322.12	1,601.00
SARIMA(1,1,0)(1,0,1)[12]	1,037.09	1,198.84

In this specific test window, the Seasonal naïve benchmark attains the lowest MAE and RMSE. This reflects the fact that the 2024–2025 period remained relatively close to the previous year's levels in several months, which favours a lag-12 benchmark. However, relying on a single split can be misleading when the underlying level is drifting downward.

5.3 Expanding-window cross-validation results

Model	Average MAE (GWh)	Average RMSE (GWh)
ETS (Holt–Winters additive)	924.65	1,144.07
Seasonal naïve	1,274.55	1,580.44
SARIMA(1,1,0)(1,0,1)[12]	1,288.78	1,462.21
ARIMA(1,1,0)	1,757.21	1,987.47

Across five folds, ETS achieves the lowest MAE and RMSE, improving MAE by roughly 27% relative to the seasonal naïve benchmark. This indicates that modelling the evolving level/trend in addition to seasonality leads to more reliable performance across forecast origins. ARIMA(1,1,0) performs worst because it does not encode the strong annual seasonal pattern.

5.4 Recommended model

Based on cross-validation, the preferred model is the additive Holt–Winters ETS specification. It balances interpretability and accuracy: it captures (i) stable annual seasonality and (ii) the gradual post-2022 decline in demand. While SARIMA also models seasonality, its cross-validated MAE is higher and its prediction intervals are substantially wider, reflecting greater parameter uncertainty and sensitivity to shocks.

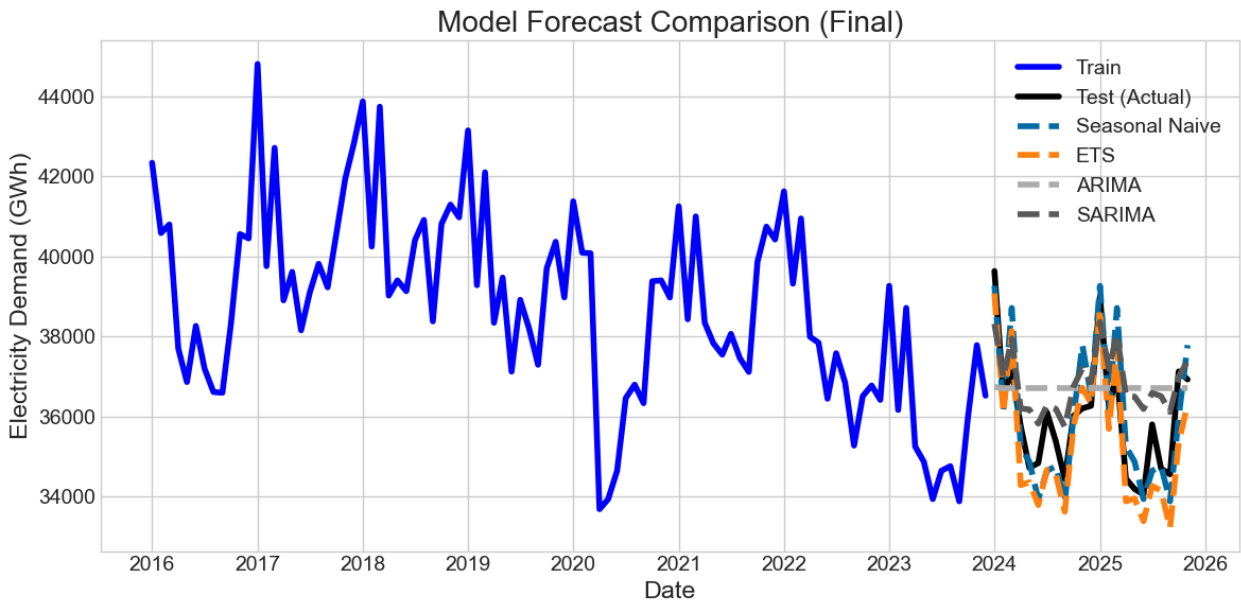


Figure 4. Out-of-sample forecast comparison on the hold-out period (Jan 2024–Nov 2025).

6. Forecast Uncertainty

Forecasts should be communicated with uncertainty, because planning decisions depend not only on the point forecast but also on the plausible range of outcomes. 95% prediction intervals were constructed for the two best-performing models (ETS and SARIMA) to illustrate uncertainty under different modelling assumptions.

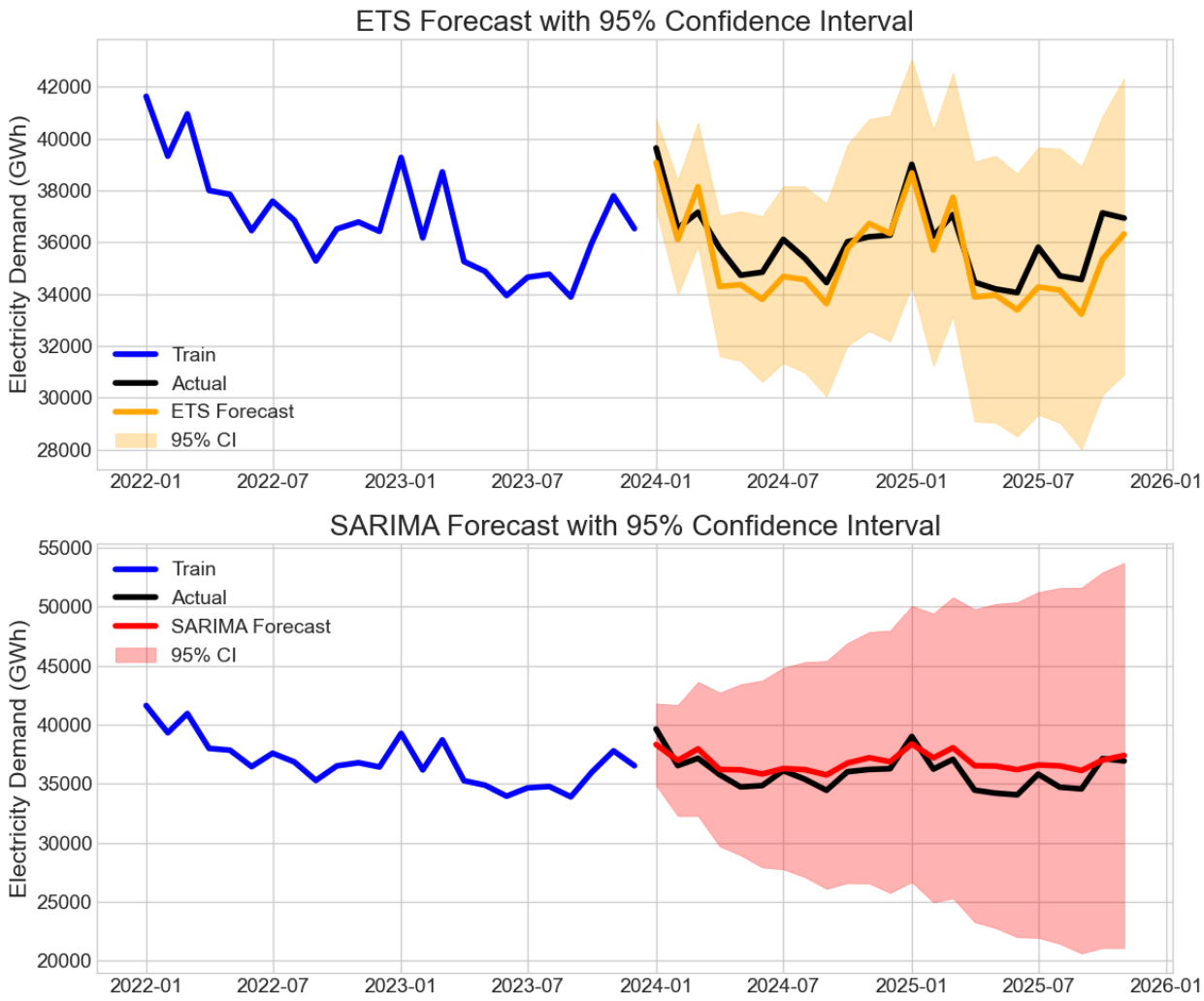


Figure 5. 95% prediction intervals for ETS (simulation-based) and SARIMA (analytic) forecasts.

For ETS, intervals were generated via simulation (1,000 replications) of the fitted additive-error state space model, avoiding strict normality assumptions. The interval width widens with the forecast horizon and remains aligned with the historical variability of the series. For SARIMA, analytic intervals widen more rapidly, reflecting accumulated forecast error variance and the model's sensitivity to structural shocks (e.g., 2020–2022). In practice, the ETS intervals are more stable and operationally useful for medium-horizon planning.

7. Limitations and Further Work

A key limitation of the Holt–Winters ETS model is its assumption that the level, trend, and seasonal pattern evolve smoothly over time. Germany's electricity demand contains disruptive events (COVID-19 in 2020 and the 2022 energy crisis) that can create abrupt, non-smooth shifts. After a sudden break, ETS adapts only gradually as new observations enter the exponential updates, which can produce biased forecasts for several months.

The preferred model may perform poorly when (i) there is a rapid regime change in industrial activity or electricity prices, (ii) weather-driven demand deviates substantially from historical patterns, or (iii) policy interventions (e.g., emergency demand reduction, subsidies, or market design changes) alter consumption behaviour. In such cases, a purely univariate model is limited because it cannot condition on exogenous drivers.

Forecasting performance could be improved by incorporating explanatory variables (ARIMAX / dynamic regression) such as heating degree days (temperature), industrial production indices, electricity prices, or renewable generation availability. A formal structural-break check (e.g., change-point detection) and regime-aware models (e.g., ETS with damped trend, SARIMAX, or state-space models with time-varying parameters) could also be explored. Finally, evaluating additional horizons (3, 6, and 18 months) would align the model choice more closely with the specific planning decisions of grid operators and traders.

AI Declaration

Generative AI tools were used in a limited and supportive capacity during this assignment. Specifically, AI assistance was sought for two purposes: debugging Python code (resolving errors in model fitting and cross-validation implementation) and supporting understanding of forecast uncertainty concepts, particularly the interpretation of prediction interval construction for ETS and SARIMA models.

All modelling decisions, parameter selections, interpretations, and written analysis are our own. The dataset, model specifications, and conclusions reflect the independent judgement of the group and not AI-generated output.